



Subtyping Speech Errors in Childhood Speech Sound Disorders with Acoustic-to-Articulatory Speech Inversion

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Abstract

Speech inversion holds much potential to describe speech errors in childhood speech sound disorders. However, the clinical interpretability of speech inversion tract variables is unknown. This study is the first to show, through linear mixed modeling, that acoustic-to-articulatory speech inversion can quantify statistically significant articulatory differences between several perceptually salient subtypes of /ɹ/ and /s/ speech sound errors, and correct /ɹ/ and /s/ targets, in American English.

Index Terms: acoustic-to-articulatory speech inversion, child speech science, speech sound disorders

1. Introduction and Motivation

1.1. Articulatory feedback for speech sound disorders

Childhood speech sound disorders reflect neurodevelopmental impairments in speech sound production [1]. Errors persisting past age 8 are thought to reflect an articulatory difficulty with underlying somatosensory or auditory-perceptual targets, most frequently impacting rhotics (i.e., /ɹ/) and sibilants (e.g., /s/) in American English. During speech therapy, clinicians often provide two types of motor learning feedback [2]. *Knowledge of results* motor learning feedback reflects a binary “correct”/“not quite” summary of the movement outcome. In contrast, *knowledge of performance* feedback provides corrective articulatory instructions (e.g., “raise your tongue blade” for /ɹ/). Knowledge of performance feedback is important for articulatory speech sound errors, in which subtle articulatory differences may result in perceptually distinct *speech sound error subtypes*. This study explores if acoustic-to-articulatory speech inversion can quantify clinically interpretable information about speech sound error subtypes produced by children and adolescents with speech sound disorders.

1.2. Acoustic-to-articulatory speech inversion

Speech inversion is the process of inferring articulatory movements from the speech signal, allowing for kinematic speech analyses in datasets that do not have a kinematic ground truth [3]. The Espy-Wilson Speech Inversion System [4, 5] is designed to predict the *degree* and *location* of vocal tract constrictions in the Articulatory Phonology framework [6], which normalizes speaker differences in vocal tract size [7] to promote speaker independent systems. Vocal tract constrictions in the Articulatory Phonology framework are modeled by six vocal tract variables: lip aperture (LA), lip protrusion (LP), tongue tip constriction location (TTCL), tongue tip constriction

degree (TTCD), tongue body constriction location (TBCL), and tongue body constriction degree (TBCD). Research has demonstrated that incorporating *source information* (fundamental frequency, F₀; periodicity, PER; and aperiodicity, APER) improves speech inversion inference [5, 8].

2. Related Works and Contributions

2.1. Speech inversion tract variable constrictions

Geometric transformations convert the ground truth kinematic pellet locations to Articulatory Phonology tract variables [9, 10]. For this study, we considered only tongue tract variables and also inverted tract variable signs relative to [10], so positive values reflected more anterior and tighter constrictions. TTCD is the minimum distance between pellet T1 and the maxilla trace. TTCL is the horizontal offset of pellet T1 from the incisor origin at the point of tightest TTCD. TBCD is the minimum distance between an arc through pellets T2-T4 and the trace of the maxilla/velum/pharynx. TBCL is the horizontal offset of the T2-T4 arc from the incisor origin at the point of tightest TBCD.

2.2. Speech sound error subtypes

2.2.1. American English /ɹ/

American English /ɹ/ involves a complex vocal tract configuration that can be categorized as “bunched” or “retroflexed.” Both variations share features like pharyngeal narrowing and lateral bracing of the tongue against the upper molars, but differ in tongue tip elevation relative to the tongue body. For “derhotic” /ɹ/ productions, midsagittal imaging may show a too-low tongue tip or blade, a too-high posterior tongue dorsum, and/or limited tongue root retraction [11]. Coronal imaging may reveal insufficient lateral bracing. Perceptually, derhotic /ɹ/ can have characteristics of other approximant sounds (i.e., /w/, /l/, or /ɥ/), fricatives (i.e., /ʃ/ or /ʒ/), or vowels (e.g., /ʌ/) [e.g., 12].

2.2.2. American English /s/

American English /s/ is articulated near the alveolar ridge, where coronal constriction creates a narrow passage for central airflow that is reinforced by lateral bracing of the tongue against the upper molars. For /s/ distortions, midsagittal imaging may show anterior/posterior tongue placement while coronal imaging may show insufficient lateral bracing. Distorted /s/ can have perceptual characteristics of /θ/, /l/, or /ʃ/ [13].

2.3. Speech inversion for speech sound disorders

Two prior studies show that adult-trained speech inversion neural networks (as in this study), can be sensitive to clinical child speech, despite a current lack of child speech inversion training data. Benway et al. [14] showed that adult-trained speech inversion inference can be highly sensitive to vocal tract gesture coordination differences for two subtypes of speech sound disorder, with and without childhood apraxia of speech, in 96 children as young as 7 (AUROC = .90). Source variable ablation significantly lowered performance compared to lip ablation, tongue ablation, and full models (Δ AUROC = -.19, $p < .001$), implicating source-filter coordination in distinguishing speech sound disorder subtypes. Benway et al. [15] showed speech inversion feature sets (median AUROC = .90) increase accuracy of binary classifiers over formants alone (median AUROC = .86) for every speaker in the [16] clinical trial.

2.4. Hypotheses and contributions

It is unknown if speech inversion estimates are sensitive to vocal tract gesture error subtypes for /ɹ/ and /s/, so this study explored if tract variable values differ between error subtypes and perceptually correct sounds. For example, articulatory phonetics [17] posits that dentalized /s/ should have a more anterior TTCL than correct /s/. For this preliminary report, one *a priori* articulatory hypothesis (Table 1) was selected for each speech sound. A paired phoneme whose articulation was theoretically related to the speech error subtype was chosen to aid interpretation. The rationale was that if a hypothesis was not supported (i.e., if dentalized /s/ was found to have more posterior TTCL values than correct /s/), examining patterns in interdental /θ/ tract variables relative to dentalized /s/ might clarify if unexpected results reflect speech inversion validity for the task. Hypotheses were constructed based on articulatory phonetics conventions and our team’s familiarity with clinical ultrasound tongue imaging for /ɹ/ and /s/ biofeedback. These hypotheses were analyzed through estimated marginal means from linear mixed-effects models. Our contribution shows that speech inversion, trained on adult ground-truth kinematic data, quantifies clinically interpretable information about tongue constriction and location for the investigated articulator and speech error subtype. The rest of this preliminary report is organized as follows: Section 3 describes the child speech dataset and perceptual labeling process, then adult speech inversion training and testing. Section 4 analyzes the child speech research questions and hypotheses using the inferences from the adult speech inversion model described in Section 3.

Table 1: *A priori hypotheses*

Sound	Error Subtype	Paired Phoneme(s)	Hypothesis vs. Correct Sound
/ɹ/	/ɹ/ → /w/	/w/	TBCL more posterior
/ɹ/	/ɹ/ → vowel	/ʌ/	TTCD lower
/s/	dentalized /s/	/θ/	TTCL more posterior
/s/	palatalized /s/	/ʃ/	TTCL more posterior
/s/	lateralized /s/	/l/	TBCD lower

3. Methods

3.1. Dataset

Data come from several IRB-approved clinical trials, longitudinal studies, and cross-sectional studies [18–26]. These studies included 118 children and 3 adults aged 3.5–45 years,

with most speakers between the ages of 6–17 years ($\bar{x} = 10.5$, $\sigma = 4.0$). Speakers contributed an average of 134 audio files each. Speech audio (words and phrases) were collected through wordlist recordings or during speech therapy activities. Audio was encoded as 44.1 kHz lossless PCM 16-bit audio in .wav containers, and was downsampled to 16 kHz during the study as required by the speech processing tools.

3.1.1. Ground Truth Labels

Error subtype ground truth is taken from expert clinician perceptual ratings with a custom Likert rating scale (Figure 1). Raters were masked to participant information and timepoint of data collection (e.g., pre-treatment, post-treatment). Because of non-perfect agreement among clinician raters, several raters rated each file. Not visible in Figure 1 are options to rate the subtype of deletion or distortion. For /ɹ/, possible subtypes included: /w/-error, /l/-error, vowel error, omitted, and other. For /s/, possible subtypes include: dentalized, lateralized, palatalized, affricated, stopped, omitted, and other.

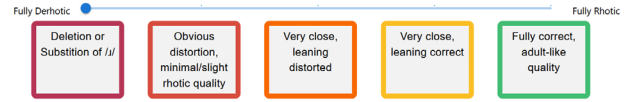


Figure 1: *Sample rating scale (for /ɹ/). Levels scored as 1–5.*

A total of 56,929 existing ratings for 17,083 candidate audio files from 137 speakers were analyzed for this study. 1,972 files were excluded from the analysis because raters indicated issues that would impact forced alignment validity (e.g., unusable audio, speech other than the target utterance in the recording, or omitted error subtypes). 5,961 audio files were retained in the analysis: 3,256 for /ɹ/ and 2,431 for /s/. These utterances met one of the following criteria: (1) /ɹ/ or /s/ was unanimously rated as correct, or (2) a majority (> 2) of raters agreed on an error subtype. Error subtypes with fewer than 15 files were excluded from the analysis (/ɹ/ → /l/, “other” /t/ errors, stopped /s/ errors, affricated /s/ errors, and “other” /s/ errors). The study’s 17,083 candidate audio files were also searched for paired phonemes /w/, /l/, /θ/, /ʃ/, and /ʌ/ (Table 1, Table 2), which were assumed to be correctly articulated. With these paired phonemes, a total of 8,608 audio files were analyzed (Table 2).

Table 2: *Distribution of error subtypes and paired phonemes.*

Sound	Ground Truth	Rating (Mean, STD)	Count
/ɹ/	correct	5 (0)	1,719
/ɹ/	/ɹ/ → /w/	1.5 (.6)	591
/ɹ/	/ɹ/ → vowel	1.5 (.6)	946
/s/	correct	5 (0)	1,401
/s/	dentalized	2.4 (.8)	604
/s/	lateralized	2.2 (.7)	272
/s/	palatalized	2.6 (.9)	154
/w/	correct	N/A	429
/θ/	correct	N/A	687
/l/	correct	N/A	167
/ʃ/	correct	N/A	636
/ʌ/	correct	N/A	1,002

3.1.2. Forced Alignment

The 17,083 candidate audio files had phone boundaries placed with utterances using the Montreal Forced Aligner version 3.0.2. Custom child speech acoustic models were used for

alignment, which preliminary work has shown improve automated speech segmentation in child speech over models trained on adult (i.e., Librispeech) data. The starting and ending frame associated with the phones of interest were used to isolate relevant parts of the tract variable time series.

3.2. Speech Inversion Model and Tract Variable Inference

3.2.1 Speech Inversion Training Data

No child data is currently available to train child speech inversion models, so the model in this study was based on the University of Wisconsin X-ray Microbeam (XRMB) dataset [9], containing speech audio and associated lip/tongue pellet data from 21 adult males and 25 adult females). We employed data reconstruction techniques [27] to increase the total dataset size from 4 to 5.3 hours. Speakers were partitioned into the same training set (36 speakers), development set (5 speakers), and test set (5 speakers) as [10]. Pellet X-Y coordinates were transformed into the tract variables described prior. Tract variable values were scaled between $[-1, 1]$ using speaker-level min-max normalization. Ground truths for the source variables were obtained using the APP detector algorithm [8].

3.2.2 Speech Inversion Architecture

This study’s speech inversion system (Figure 2) uses the self-supervised WavLM pretrained model [28] to extract speech features. The 25-layer WavLM output was processed through two convolutional layers, first reducing it to 16 layers then a single-layer representation. This was followed by two gated recurrent unit layers with 256 and 128 units. To match the target 100 Hz sampling rate, we upsampled the WavLM 50 Hz output by a factor of two. The final feature set was passed through two dense layers (128 and 9 units), mapping to six tract and three source variables. We optimized the model using Adam with a plateau learning rate scheduler, starting at $5e-4$, and determined an optimal batch size of 8 via grid search. To prevent overfitting, we applied early stopping with a patience of 10 epochs. A weighted combination of Pearson Product Moment Correlation (PPMC) and root mean squared error served as the loss function.

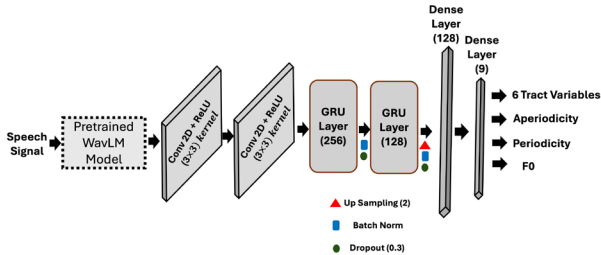


Figure 2. Speech inversion model architecture.

3.2.3 Speech Inversion Performance on XRMB Test Set

We utilized the same XRMB test set as [10] to evaluate the PPMC performance of the proposed speech inversion system for adult speech. As presented in Table 3, the average PPMC correlation across all six tract variables was 0.85, representing a 3.93% relative improvement for the same speakers compared to the speech inversion system proposed by [10]. Key differences in the present model are the use of WavLM (vs. HuBERT) and the estimation of glottal source tract variables PER (.93), APER (.88), and F_0 (.76), which were not examined further in this study. These results demonstrate the efficacy of

the proposed speech inversion system in enhancing speaker-independent performance for tract variable estimation.

Table 3: Speech inversion performance (PPMC). The average includes three source variables described parenthetically.

Metric	LA	LP	TTCL	TTCD	TBCL	TBCD	Average
PPMC	.92	.76	.84	.94	.77	.85	.85

3.2.4 Tract Variable Inference

The adult-trained speech inversion model was used to infer the tract variable values for the child speech data set. We only examined tongue tract variables for the purposes of this report, by calculating the average of each tract variable time series within the force-aligned interval.

3.3. Statistical Analysis of Tract Variable Estimates

Linear mixed models examined if tract variable values for speech error subtypes differed from correct speech sound in interpretable ways according to our *a priori* hypotheses. Two separate linear mixed models (one /ɪ/, one /s/) were fit to examine the interaction of *tract variable channel* and *speech sound* on the outcome, *tract variable value*. A random intercept was included to model speaker specific variation, with repeated audio files clustered within speakers. Models were fit with *lme4* and estimated marginal means were calculated with *emmeans*, both in R. Reverse pairwise contrasts were applied to compare the tract variable values for error subtypes and paired phonemes against the tract variable values for correct productions. Both linear mixed models converged without fit warnings.

4. Results

The null hypothesis, that there was no difference in tract variable values for error subtypes vs. correct sounds, was rejected for four of our *a priori* hypotheses (Table 4), providing evidence for each of our hypotheses except for lateralized /s/.

Table 4: Results of estimated marginal means comparisons. Evidence supporting our hypotheses is bolded.

Error Subtype	Estimate	<i>p</i>	Interpretation vs. Correct
/ɪ/ → /w/	-0.19	< .001	Tongue body more posterior
/ɪ/ → vowel	-0.14	< .001	Tongue tip lower
dentalized /s/	0.05	< .001	Tongue tip more anterior
palatalized /s/	-1.10	< .001	Tongue tip more posterior
lateralized /s/	0.11	< .001	Tongue body higher

We hypothesized /ɪ/ → /w/ errors (Figure 3) would have an average tongue body constriction that was more posterior (i.e., more velar) than correct /ɪ/. Estimated marginal means support this hypothesis, indicating that group-average normalized TBCL was significantly decreased in /ɪ/ → /w/ errors compared to correct /ɪ/ (-0.19, SE= 0.006, z-ratio = -29.4, $p < .0001$). The same pattern was seen for paired /w/ phonemes, which had significantly more posterior TBCL than /ɪ/ → /w/ errors (-0.05, SE= 0.008, z-ratio = -6.6, $p < .0001$).

We hypothesized that /ɪ/ → vowel errors would result in an average tongue tip constriction that was lower than correct /ɪ/. Estimated marginal means support this hypothesis, indicating that group-average normalized TTCD was significantly decreased in /ɪ/ → vowel compared to correct /ɪ/ (-0.14, SE=0.005, z-ratio = -24.8, $p < .0001$). Paired /ɪ/ had a

significantly lower TTCD than vowel errors (-0.02 , $SE = 0.006$, $z\text{-ratio} = -2.9$, $p = 0.027$).

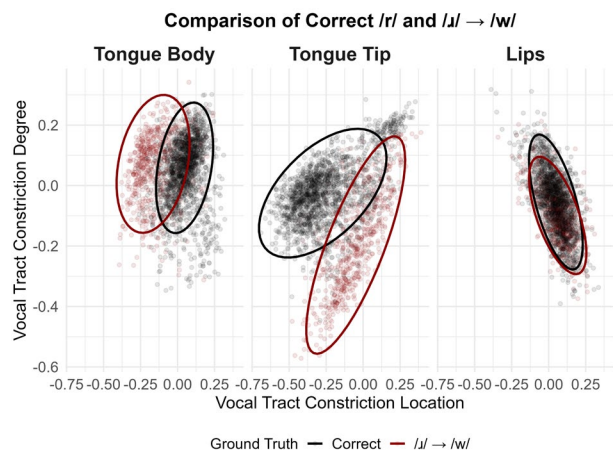


Figure 3. Distribution of tract variables for $/r/ \rightarrow /w/$ errors (red) vs. correct $/r/$ (black). Ellipses = 95% confidence regions.

We hypothesized that the average TTCL in dentalized $/s/$ would be more anterior than in correct $/s/$. Estimated marginal means support this hypothesis, indicating that group-average normalized TTCL was significantly increased in dentalized $/s/$ versus correct $/s/$ (.051, $SE=0.006$ $z\text{-ratio}= 9.1$, $p < .0001$). Paired $/\theta/$ had significantly more anterior TTCL than dentalized $/s/$ errors (.022, $SE = 0.006$, $z\text{-ratio} = 3.4$, $p = 0.01$).

We hypothesized that the average TTCL in palatalized $/s/$ would be more posterior than in correct $/s/$. Estimated marginal means support this hypothesis, indicating that group-average normalized TTCL was significantly decreased in palatalized $/s/$ versus correct $/s/$ (-1.10, $SE= 0.009$, $z\text{-ratio} = -12.3$, $p < .0001$). Paired $/j/$ had significantly more posterior TTCL than palatalized $/s/$ (-.046, $SE = 0.01$, $z\text{-ratio} = -4.8$, $p < .0001$).

We hypothesized that the average TBCD in lateralized $/s/$ would be lower than in correct $/s/$. Estimated marginal means did not support this hypothesis; group-average normalized TBCD was significantly increased (i.e., tighter) in lateralized $/s/$ compared to correct $/s/$ (.108, $SE = 0.007$, $z\text{-ratio} = 14.7$, $p < .0001$). The opposite TBCD pattern was seen for word-initial (i.e., alveolar) $/l/$ compared to correct $/s/$ (-.06, $SE = 0.009$, $z\text{-ratio} = -6.9$, $p < .0001$). TBCD of $/l/$ was significantly lower than lateralized $/s/$ (-.17, $SE = 0.01$, $z\text{-ratio} = -16.2$, $p < .0001$).

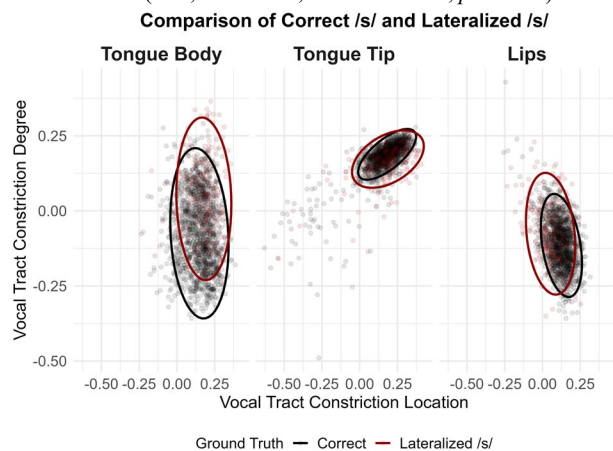


Figure 4. Distribution of tract variables for lateralized $/s/$ (red) vs. correct $/s/$ (black). Ellipses = 95% confidence regions.

5. Discussion

The goal of this study was to explore if acoustic-to-articulatory speech inversion quantified clinically interpretable information about speech sounds produced by children and adolescents with speech sound disorders. The long-term goal of this line of work is to motivate speech inversion tract variable inference as clinically interpretable speech measures for machine learning.

Our study shows that, for four of our five *a priori* hypotheses, adult-trained speech inversion data provides clinically intuitive articulatory information about child speech that aligns with articulatory phonetic conventions [17]. Speech inversion estimates indicated that $/r/ \rightarrow /w/$ errors had a more posterior tongue body than correct $/r/$, $/l/ \rightarrow$ vowel errors had a lower tongue tip than correct $/l/$, dentalized $/s/$ had a more anterior tongue tip than correct $/s/$, and palatalized $/s/$ had a more posterior tongue tip than correct $/s/$. Furthermore, evidence from paired phoneme tract variables supports the common clinical notion that $/r/ \rightarrow /w/$ errors reflect intermediary gestural and perceptual elements between $/r/$ and $/w/$, $/l/ \rightarrow$ vowel errors are intermediary to $/l/$ and $/\Lambda/$, dentalized $/s/$ errors are intermediary to $/s/$ and $/\theta/$, and palatalized $/s/$ errors are intermediary to $/s/$ and $/j/$. It is notable that these articulatory differences were noted even using speech signal processing methods that downsample audio to 16 kHz, given that child fricatives can exceed 8 kHz [29].

While TBCD was significantly different in lateralized $/s/$ versus correct $/s/$, the effect was in the opposite direction than we hypothesized. The intuition driving our hypothesis was that lateral bracing supports central airflow in correct $/s/$, lateral $/s/$ airflow reflects insufficient lateral bracing, and insufficient lateral bracing would be reflected by a lower tongue body along the midsagittal plane than a tongue with sufficient lateral bracing. Without a phonemic $/l/$ in American English, $/l/$ was selected as a paired phoneme. Given that speech inversion suitably captured the articulation of other phonemes, the results for lateral $/s/$ may indicate that there are more complex tongue dynamics at work for lateralized $/s/$, that midsagittal speech inversion might not sufficiently capture lateral tongue gestures, and/or that $/l/$ is not a suitable analogy for lateralized $/s/$. Our ongoing work investigates each of these possibilities.

Our ongoing work also expands upon the study reported here. We are examining additional tract variable hypotheses for each speech error subtype. The analysis is also expanding to account for the simultaneous impact of constriction and degree. The need to examine both simultaneously is highlighted by the simultaneous difference in constriction location and degree evident in the Tongue Tip panel of Figure 3. This same figure panel also appears to show two clusters in correct $/r/$, possibly representing bunched and retroflexed $/r/$, and our future work aims to explore this further. It is also possible that using time-averaged tract variables has obscured important time series information, which our future will address. Finally, this study leaves an open question of how speech inversion tract variable inference would differ if they were measured by a speech inversion system validated for child speech, which is a direct goal of our current research.

6. Conclusions

Acoustic-to-articulatory speech inversion, trained on adult ground-truth kinematic data, quantifies clinically interpretable information about tongue constriction and location for $/r/ \rightarrow /w/$, $/l/ \rightarrow$ vowel, dentalized $/s/$, and palatalized $/s/$ speech sound errors in children and adolescents with speech sound disorders.

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